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LearnNova: Automated Generation of Educational Content Using LLMs

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Chapter 1 Literature Review / Related Work

This chapter presents the detailed literature review of studies relevant to video generation, manim code generation and training of domain specific LLM. A summary table is provided at the end to summarize the findings.

1.1 Definitions, Acronyms, and Abbreviations

To ensure a clear understanding of the terminology used throughout the chapter, a list of acronyms and their respective meanings are provided in the list below: **LLM:** Large Language Model

AI: Artificial Intelligence

TTS: Text-to-Speech

Diffusion Model: Probabilistic generative model that iteratively refines noise into structured data

Transformer: Deep learning architecture based on self-attention

RAG: Retrieval-Augmented Generation

API: Application Programming Interface

SDG: Sustainable Development Goal

Manim: Mathematical Animation Engine

FYP: Final Year Project

1.2 Detailed Literature Review

This section offers a comprehensive review of studies that are relevant to our project. It details the techniques applied, the results obtained, and provides a critical analysis of these studies. The relevance of these studies to our project is also discussed.

1.2.1 Video Generation

Recent advancements in video generation have been largely driven by the evolution of diffusion and transformer-based generative models. Researchers have introduced frameworks capable of producing short, high-quality video clips directly from textual prompts, marking a significant leap in generative artificial intelligence. One of the early approaches, AnimateDiff [1], extended diffusion-based image models toward animation by inserting motion modules within Stable Diffusion pipelines. This modular design enabled consistent short animations from text while preserving visual fidelity.

In the article the authors address a core limitation in current text-to-image diffusion models: the inabil-

ity to generate animations while maintaining the visual quality and personalization of the base models. The paper introduces AnimateDiff[1], a plug-and-play framework that adds motion capabilities to any personalized text-to-image model, such as Stable Diffusion, without requiring model-specific fine-tuning. The system includes a motion module that learns general motion dynamics from real-world video datasets, and a domain adapter that ensures stable visual quality by bridging the gap between video data and static image models. Additionally, the authors propose MotionLoRA, a lightweight fine-tuning technique that allows the model to learn specific motion styles such as panning, zooming, or rolling with minimal data and computational cost. Through this modular setup, AnimateDiff[1] enables personalized image models to generate coherent, temporally smooth animations while preserving their original appearance and quality.

When connecting this work to the LearnNova, there are several shared design philosophies. Both systems emphasize modular architectures that separate distinct processes for example in AnimateDiff[1], this involves separating motion learning from visual learning, while in LearnNova, it involves separating script generation, Manim code generation, audio narration, and video rendering. AnimateDiff's plug-and-play motion module aligns conceptually with LearnNova's goal of developing reusable components for synchronized animation and narration. Furthermore, the MotionLoRA fine-tuning approach mirrors LearnNova's strategy of using fine-tuned LLMs to adapt general AI models for specialized educational tasks. AnimateDiff's success in producing consistent, controllable motion without retraining large models supports LearnNova's vision of automating the creation of visually engaging, accurate, and synchronized educational videos efficiently. Ultimately, while AnimateDiff focuses on general video generation from diffusion models, its principles of motion modularity, fine-tuning efficiency, and domain preservation directly inspire LearnNova's approach to scalable and intelligent educational content generation.

Building on similar principles, Chen et al. introduced VideoCrafter and its successor, VideoCrafter2 [2], which addressed dataset limitations through spatio-temporal adapters and multi-stage training. These models achieved high-resolution generation with improved text-video alignment, becoming foundational open frameworks for future video synthesis systems

In article titled VideoCrafter2[2] explains a key problem in text-to-video generation. The problem is how to create videos that look good and move smoothly without using very large and organized video datasets. Commercial tools such as Runway Gen-2 and Pika Labs create realistic videos. However, these systems depend on private and large video collections that the research community cannot access. It tried to solve this problem by using a method that trains diffusion-based video models with only low-quality video data and high-quality synthetic images. The goal is to make better videos without using

expensive data. The method separates motion learning and appearance learning at the data level. Motion is learned from low-quality videos such as WebVid-10M. Appearance and picture quality are learned from high-quality synthetic images produced by models such as Midjourney. By keeping these two types of learning separate, VideoCrafter2[2] achieves good motion and clear visuals without needing large video resources.

The link between space and time in video diffusion models was also studied in VideoCrafter2[2]. It shows that when a model learns spatial and temporal modules together, it becomes stronger and more stable during fine-tuning. This helps the model keep motion smooth even when the visual quality is improved. The researchers used LoRA-based fine-tuning to adjust the spatial modules with high-quality images. This approach improves visual realism while keeping motion consistent. The team tests their model using the EvalCrafter benchmark. The results show that the model performs well compared to top commercial systems. It does this in terms of visual quality, match between text and video, and user preference, even though it uses much less data.

When we connect this work to the LearnNova project, we can see many common ideas. LearnNova is an AI system that creates educational videos in subjects such as mathematics and physics. It uses large language models, Manim animations, and text-to-speech tools to do this automatically. Both VideoCrafter2[2] and LearnNova try to overcome limits found in existing generative systems. They both use fine-tuning, modular structures, and data-efficient training. LearnNova focuses on a teaching problem, it must create accurate, synchronized, and clear educational videos without needing programming skills, where as, VideoCrafter2[2] focuses on smooth motion and clear visuals in general video generation. . Moreover, both systems use LoRA or QLoRA fine-tuning to adapt pre-trained models for special purposes.

Parallel to these diffusion-based innovations, CogVideoX model[3] introduced a large-scale text-to-video generation framework designed to improve the efficiency, quality, and accessibility of AI-based video creation. The key objective of CogVideoX is to generate high-resolution, temporally consistent, and semantically accurate videos from textual descriptions using compressed fine-tuning and efficient diffusion-based architectures.

Its purpose aligns closely with LearnNova, which also focuses on automating AI-generated educational video creation. Both aim to produce high-quality, context-aware, and synchronized visual content based on textual input.

CogVideoX[3] employs a modular architecture combining a multimodal text encoder, a video diffusion model, and a video autoencoder (VAE) that compresses video representations for efficient storage and fine-tuning. Through compressed fine-tuning, the model adapts quickly to new tasks and domains,

reducing computational overhead while maintaining output fidelity. Its Diffusion Transformer (DiT) backbone ensures long-range temporal consistency, allowing it to capture smooth motion and rich visual detail across frames.

Training on large-scale, curated datasets with structured captions, CogVideoX[3] achieves strong results across benchmarks, outperforming previous text-to-video systems in realism, motion fluidity, and semantic alignment. Its performance validates that diffusion-based generative frameworks can create coherent, high-resolution videos from simple text inputs.

In relation to LearnNova, CogVideoX[3] provides conceptual and architectural parallels. Both frameworks rely on text-based prompts to generate synchronized video content. Where CogVideoX[3] generates cinematic scenes from descriptive prompts, LearnNova generates educational animations using script generation, Manim-based code rendering, and synchronized narration. Both systems emphasize automation, accuracy, and multimodal integration to ensure high-quality video output.

Furthermore, CogVideoX's use of compressed fine-tuning and efficient training directly informs LearnNova's approach to developing a self-hosted fine-tuned LLM for generating reliable Manim code. Both share a commitment to reducing computational costs while maintaining scalability and precision.

In conclusion, CogVideoX[3] demonstrates how advanced multimodal generation models can transform text inputs into rich, temporally coherent videos which is a principle that directly supports LearnNova's mission of automating AI-generated educational video creation. The integration of text, animation, and audio in both systems highlights the growing convergence between AI-driven video generation and intelligent educational content creation, paving the way for scalable, high-quality, and accessible learning experiences.

Another key milestone in this domain is Tencent's HunyuanVideo [4] proposed a multimodal framework integrating visual, textual, and auditory modalities, resulting in stable long-duration motion generation. Both works exemplify a transition toward general-purpose, high-fidelity video generators capable of producing complex scenes, though their emphasis remains on cinematic realism rather than educational visualization.

HunyuanVideo[4] presents an open-source video foundation model designed to rival closed-source systems in quality and performance. It combines advanced data curation, large-scale diffusion architectures, and multimodal learning to create realistic and semantically aligned videos from text prompts.

HunyuanVideo[4] follows a structured pipeline that includes comprehensive data filtering, structured annotation, and progressive pre-training using both images and videos. The model architecture integrates a 3D Variational Autoencoder for compressing visual data and a Transformer-based diffusion model for

generating dynamic video sequences. Text prompts are processed by a Multimodal Large Language Model (MLLM) that ensures strong alignment between language and visuals.

The system's training process follows a progressive curriculum, starting from simple low-resolution clips to high-quality, long-duration videos, achieving a balance between efficiency and fidelity. With over 13 billion parameters, HunyuanVideo[4] delivers remarkable motion quality, text-video alignment, and scene realism, outperforming several commercial models like Runway Gen-3 and Luma 1.6. Its structured captions, camera movement recognition, and multimodal integration enable the model to produce coherent, context-rich, and high-resolution videos.

The framework extends to multiple downstream applications, including video-to-audio generation, image-to-video transformation, and avatar animation, emphasizing the creation of synchronized, expressive, and immersive multimedia content.

In relation to LearnNova, both systems share the core objective of automating content creation through AI-based multimodal generation. While HunyuanVideo[4] focuses on video synthesis and understanding, LearnNova focuses on educational video generation, automating the entire process from script creation (via Gemini API) to animation generation (via fine-tuned Manim LLM) and audio synchronization (via TTS). Both frameworks emphasize accuracy, synchronization, and automation, demonstrating how multimodal AI can enhance content creation and accessibility.

Thus, HunyuanVideo[4] provides strong conceptual and technical foundations for LearnNova, showcasing how advanced multimodal architectures, structured data pipelines, and text-video alignment techniques can be adapted to create AI-driven educational videos that are visually coherent, contextually precise, and pedagogically effective.

Collectively, these advancements demonstrate remarkable progress in video generation, showcasing strong potential for automated visual content creation. The cited papers collectively represent the technological foundation and motivation for the LearnNova project. Works like AnimateDiff [1], VideoCrafter2 [2], demonstrate how diffusion-based architectures can generate short, high-quality videos but still struggle with temporal consistency—an essential feature for educational tutorials. CogVideoX [3] introduce transformer-based text-to-video generation with stronger semantic understanding, showing potential for script-driven animation but prioritizing cinematic realism over pedagogical clarity. Similarly, HunyuanVideo [4] explore multimodal and large-scale video generation, aligning conceptually with LearnNova's integration of text, visuals, and audio, though they lack fine-grained control for concept explanation. Open and community-driven systems emphasize accessibility, modularity, and creative freedom—principles central to LearnNova's vision of democratizing educational video creation for non-technical users. While these models focus on realism, storytelling, or general-purpose genera-

tion, LearnNova extends this progress toward domain-specific, pedagogically accurate, and automated Manim-based animation, filling a critical gap in AI-driven educational content generation.

The gap identified through this review lies in the absence of domain-specific systems that prioritize conceptual accuracy, structured narration, and pedagogical intent. LearnNova addresses this gap by integrating large language models for script generation, specialized fine-tuning for Manim-based animation, and synchronized text-to-speech narration. This approach combines automation, accessibility, and domain alignment to generate precise and visually coherent educational videos in mathematics and physics.

1.2.2 Manim Code Generation

This section have explored the automation of educational content creation through multimodal systems combining text, code, and video generation. Li et al. [5] introduced Theorem Explain Agent, an agentic framework that employs large language models (LLMs) to generate Manim-based animations for theorem proofs. Their approach effectively combined symbolic reasoning and visualization, demonstrating the potential of LLMs to transform abstract mathematical logic into comprehensible animated explanations. In this the framework combine both visual and textual reasoning. The study extends the scope of AI interpretability by joining large language models, agent planning, and automated visualization through Manim animations. The system tries to connect symbolic reasoning with human understanding by offering structured and multimodal explanations across various STEM fields.

TheoremExplainAgent[5] has two large language model agents that work together. The first is a planner agent that builds the narrative and storyboard. The second is a coding agent that changes the plan into executable Manim scripts. This structure reflects current ideas in agent-based AI systems, where models can plan and perform tasks by using tools and interacting with their environments. The study also includes a retrieval augmented generation method. It allows the agents to access documents, visual examples, and code templates during generation to improve animation quality. By using these methods together, TheoremExplainAgent shows that structured video generation can be a clear way to explain complex theorems.

While TheoremExplainAgent[5] focuses on multimodal theorem explanation, our system extends the same philosophy to general educational video generation, aiming to produce 3Blue1Brown-style mathematical animations through an automated, AI-driven process. To evaluate multimodal reasoning in a structured way, the authors present TheoremExplainBench. It is a benchmark that includes 240 theorems from mathematics, physics, chemistry, and computer science. The theorems cover different levels of difficulty. The benchmark uses five main metrics for evaluation. These are accuracy and depth, vi-

sual relevance, logical flow, element layout, and visual consistency. This design helps to measure both factual and visual quality. It is similar to earlier multimodal reasoning datasets such as ScienceQA and TheoremQA. However, it focuses on multimodal explanation instead of text-only reasoning. Like TheoremExplainAgent’s two-agent system LearnNova similarly integrates modular automation. We employ the Gemini API for structured script generation and a fine-tuned LLM trained to produce Manim animation code corresponding to that script. This mirrors the planner–coder synergy found in TheoremExplainAgent but adapts it to a broader educational context beyond theorem reasoning.

Moreover, both systems leverage retrieval and reasoning mechanisms to enhance accuracy and coherence. In our setup, retrieval-like optimization occurs when the LLM accesses fine-tuned Manim templates and animation patterns, ensuring that generated visuals are pedagogically sound and mathematically correct. Similarly, the text-to-speech (TTS) integration in our pipeline parallels TheoremExplainAgent’s emphasis on multimodal synthesis—combining text, visuals, and narration to promote conceptual clarity.

In essence, TheoremExplainAgent[5] provides a strong conceptual foundation for our project, validating the idea that LLM-driven Manim generation can serve as an effective medium for AI-based education and explainability. By adapting its multimodal reasoning and agentic planning principles, our project advances toward an automated framework capable of producing accurate, synchronized, and visually engaging educational videos, thereby contributing to the broader goal of AI-assisted teaching and content creation. The results show that TheoremExplainAgent performs much better in long-form video generation than text-to-video models but some problems remain in visual layout, consistency, and information retrieval which are the aims of LearnNova to fulfil. These challenges are also seen in other studies on retrieval augmented code generation.

Similarly, Herrmann et al. [6] developed Manimator, a pipeline that converts scientific text into Manim animations, aiming to simplify the process of visualizing complex research concepts. Although the system produced visually appealing results, it lacked synchronized narration and fine-grained control over timing and educational pacing.

Manimator[6] is an open-source system that uses Large Language Models to convert complex research papers and their natural language descriptions into visual explanatory animations by using the Manim engine. The main goal of Manimator[6] is to make scientific and mathematical material easier to understand. It does this by changing difficult written content into dynamic and engaging visuals.

Manimator[6] works through a process that has three main stages. In the first stage, the system focuses on understanding the input and turning it into a structured visual plan. A multimodal large language model (LLM) analyzes the source, which can be a research paper, a textbook PDF, or a plain text prompt.

When the input document is complex, models like Gemini 2.0 Flash handle both visual and textual understanding. On the other hand, LLaMA 3.3 70B deals with text-based prompts. The model identifies and arranges important parts like definitions, equations, theorems, and conceptual metaphors. These are collected in a Markdown “Scene Description” file that lists topics, main LaTeX-formatted equations, proposed visuals, and animation flow. This stage changes detailed academic material into a clear step-by-step storyboard that can be used for visual rendering.

When the structured scene description is ready, the second stage begins. This stage converts the plan into executable Python code by using a code-specialized LLM like DeepSeek-V3 or Qwen2.5-Coder. This model works as an expert in the Manim animation engine and follows a few selected examples to maintain quality and consistency. It generates classes, scenes, and transitions with correct sequencing, timing, and clarity. The model also adds mathematical notations, makes smooth animations, and avoids visual clutter by keeping enough space between visual elements. In this way, this stage changes the conceptual descriptions into Manim-compatible scripts that show both the logical and visual flow of the content.

In the last stage, the Python script is executed through the Manim library to create an MP4 animation. Dependencies like FFmpeg manage the video compilation and ensure a smooth output. The completed animation explains the given concepts through synchronized movement, text, and graphics. This helps in understanding abstract topics more easily. Manimator’s performance was tested using the TheoremExplainBench (TEB) benchmark. It achieved the highest overall accuracy of 0.845 among similar systems. Human evaluations by engineering students also showed that Manimator’s output was clear, well-organized, and effective in simplifying complex academic material into engaging educational animations.

This study highlights that creating Manim animations usually requires strong programming and domain knowledge. The system was evaluated on the TheoremExplainBench benchmark, achieving high accuracy and positive feedback for clarity and coherence. Its success demonstrates that automatic generation of Manim-based educational animations can reduce the need for programming expertise and make visual learning more accessible.

Our project, LearnNova, builds upon this concept with an extended goal. While Manimator focuses on visualizing academic papers and theoretical material, LearnNova automates the entire process of creating educational videos. It not only generates Manim animations but also includes synchronized narration and script generation. In LearnNova, a user provides a topic prompt, which is processed through Gemini API for script creation and a fine-tuned LLM for Manim code generation. The script is converted into audio using a text-to-speech engine, and both audio and animation are merged into a final educational

video.

In summary, both systems aim to simplify learning through AI-generated visualizations. However, LearnNova advances this vision by producing complete didactic video tutorials with narration and animations, offering a more interactive and accessible form of automated educational content creation in mathematics and physics. Both systems share the goal of making visual learning more accessible, scalable, and effective through animations.

Complementary research has examined end-to-end video generation workflows. The Text2Video system [7] explored converting narrative scripts into animated video sequences using LLMs and speech synthesis, primarily for storytelling applications rather than instructional content.

The study Text2Video[7] handled the increasing demand for multimedia content and the need for efficient video creation tools have driven research into automatic video generation systems. by introducing an innovative system designed to automatically generate game-related videos directly from text scripts. The system aims to simplify video production for novice users by combining natural language processing, cross-modal retrieval, and automated video editing techniques into a cohesive workflow. The system architecture comprises five core modules: material management, natural language processing (NLP), cross-modal retrieval, text-to-speech (TTS), and video editing. The material manager handles the collection and organization of raw video, image, and audio materials. It employs a highlight extractor to identify significant in-game events such as “first blood” or “triple kill” using feature matching, OCR, and convolutional neural networks (CNNs) for hero recognition. Each material is annotated with semantic labels to enable efficient retrieval during video generation.

The Text2Video[7] system represents a significant step toward automated multimedia generation, particularly in the gaming domain. It bridges the gap between narrative scripts and video output, enabling users to produce visually and audibly rich content with minimal manual intervention. By leveraging state-of-the-art NLP, computer vision, and speech synthesis technologies, the system democratizes video production, making it accessible to non-experts while maintaining high creative quality. This research paves the way for future advancements in AI-driven storytelling and user-generated content creation.

Similar to how the Text2Video[7] automated video production from text scripts using NLP and multimedia retrieval, our project extends this concept into the educational and mathematical visualization domain.

Our proposed system automates the creation of 3Blue1Brown-style mathematical videos, integrating script generation, AI-based code synthesis, and multimedia rendering into a unified workflow. Instead of relying on manual scripting or animation, the system takes a topic prompt as input and leverages the Gemini API to generate an educational script. A custom fine-tuned LLM, trained specifically for

Manim code generation, then produces synchronized mathematical animations that visually represent the concepts described in the script.

To enhance the learning experience, the generated script is transformed into natural-sounding narration using a text-to-speech (TTS) module with precise timing alignment. The Manim-rendered visuals and the synthesized audio are then merged, producing a coherent, synchronized educational video. This automation pipeline eliminates the need for manual video editing or coding expertise, thereby making high-quality, mathematically accurate video content creation accessible to educators and students alike.

While Yu et al.'s Text2Video[7] system focused on cross-modal retrieval and game event narration, our approach emphasizes conceptual accuracy, mathematical visualization, and pedagogical quality. By combining LLM-based code generation, TTS synchronization, and Manim-based animation, the project demonstrates a novel direction in AI-driven educational media generation. It represents a significant step toward democratizing the production of explainable, visually rich, and AI-assisted learning resources.

More recently, Multi-RAG [8] proposed a comprehensive system that improves how artificial intelligence understands videos. The proposed system, Multi-RAG, combines video, audio, and text data by using the concept of Retrieval-Augmented Generation (RAG). Its main purpose is to help AI models give factual and context-based answers instead of generating assumptions.

Traditional LLMs face two main problems. They have limited access to updated data, and they sometimes produce false or imagined information. Multi-RAG solves both problems by retrieving related information from external sources before giving a response. The authors aim to design a system that is efficient, dependable, and useful for real applications like automated assistance, video summarization, and intelligent reasoning with visual content.

Large language models are good at reasoning with text. However, they face difficulty in understanding visual and audio data. Vision-language models can describe still images but cannot capture motion or change in videos. Videos contain continuous visuals, sound, and speech. All of these elements need to be understood together to achieve full comprehension.

Retrieval-Augmented Generation (RAG) helps AI combine stored knowledge with real-time reasoning. The authors extend this idea to multimodal input. They propose Multi-RAG as a system that retrieves important video and audio descriptions before generating an answer. This process lowers the model's dependence on its internal memory and helps it create accurate and well-grounded responses. The Multi-RAG system has three layers: perception, knowledge storage, and generation.

The Perception Layer extracts details from the video. It applies two frame sampling methods, uniform and change-based, to select key frames in an efficient way. A Vision-Language Model (VLM) creates

text descriptions for every frame, and Whisper converts the audio into text. These results are then joined together to make detailed segment descriptions.

The Knowledge Database stores and organizes these segment descriptions. It saves them in Markdown format, divides them into smaller overlapping parts, and embeds them by using OpenAI embeddings. The embeddings are kept in a Chroma vector database. This database allows fast and accurate retrieval of relevant information through semantic search.

The RAG Agent and LLM manage the question answering process. When a user asks a question, the system retrieves the top-k most related chunks of information. These chunks are then passed to a large language model. The system uses two prompting types. One is an open prompt that allows flexible reasoning. The other is a conservative prompt that focuses on factual accuracy.

The main strength of Multi-RAG[8] is its balanced use of multiple input types. By combining video, audio, and text, it captures deeper context than models that use only text or only images. The retrieval method connects its answers to real data, which improves factual accuracy and makes the output easier to interpret.

The framework is also efficient in using resources. Its selective frame sampling method helps reduce unnecessary computation while keeping the understanding intact. Its flexible prompting setup allows developers to choose between creative and factual responses based on their goals.

Although Multi-RAG[8] gives strong results, some limitations still exist. Handling video and audio data can cause privacy and security issues that must be solved before using the system in real applications. The model still relies heavily on LLM reasoning, so small hallucinations can still appear.

The results of Multi-RAG[8] demonstrate strong performance on benchmark datasets, achieving accuracy comparable to larger models while using fewer computational resources. Its integration of multi-modal data and retrieval mechanisms ensures grounded, fact-based outputs and a better understanding of complex visual content.

This work relates closely to our project, LearnNova, which also integrates multimodal elements for automated video generation. While Multi-RAG focuses on understanding and reasoning over existing video content, LearnNova applies similar principles of AI-based multimodal processing for video creation. Our system automates the entire process that is from topic input to script generation, animation creation using Manim, audio synthesis, and video compilation. By combining text, audio, and animation, LearnNova aims to produce synchronized, high-quality educational videos that visually and narratively explain mathematical and scientific concepts.

In summary, both Multi-RAG and LearnNova share the goal of improving AI's interaction with visual

and linguistic data. However, while Multi-RAG focuses on video comprehension through retrieval and reasoning, LearnNova focuses on video generation through automation and synchronization. Together, these approaches highlight the growing role of multimodal AI in enhancing both understanding and creation of educational content.

The referenced studies collectively outline the research landscape that LearnNova builds upon and extends. TheoremExplainAgent [5] demonstrates how large language models can generate Manim-based animations for theorem proofs, showing the feasibility of combining symbolic reasoning with visualization—a concept central to LearnNova’s goal of automating mathematical explanation through visual content. Manimator [6] similarly focuses on transforming scientific text into Manim animations, but it lacks synchronized narration and pedagogical pacing, limitations that LearnNova directly addresses through fine-grained temporal alignment and integrated text-to-speech modules. LearnNova improves upon this by fine-tuning an LLM on curated 3Blue1Brown code to ensure both syntactic and conceptual correctness. In parallel, Text2Video [7] explore automated video generation from text for narrative and educational purposes, respectively, but lack robust multimodal synchronization and domain-specific reasoning—key features LearnNova introduces for STEM-focused content. Multi-RAG [8] advances multimodal retrieval-augmented generation for video understanding, offering a framework that informs LearnNova’s multimodal design, though its focus is not on generative animation or fine-tuned mathematical visualization. Together, these works motivate LearnNova’s contribution: a domain-specialized, end-to-end automated system that integrates LLM-based scripting, pedagogical animation, and synchronized narration for high-quality educational video generation.

These studies establish the foundation for automated educational video synthesis but reveal several persistent gaps. Existing systems either focus narrowly on specific domains, such as theorem explanation, or struggle with maintaining synchronization, semantic accuracy, and pedagogical clarity in generated outputs. The proposed LearnNova project seeks to bridge these gaps by developing a fine-tuned, self-hosted LLM trained specifically on Manim code derived from 3Blue1Brown’s [9] repository. By integrating script generation via Gemini, precise text-to-speech narration with timing control, and automated rendering through Manim, LearnNova aims to produce coherent, synchronized, and pedagogically effective educational videos in mathematics and physics. This approach not only extends the multimodal frameworks established in prior work but also introduces domain specialization, quality assurance, and end-to-end automation—key steps toward scalable, AI-driven content generation for STEM education.

1.3 Literature Review Summary Table

Throughout the research, we've studied various research papers to get an in-depth knowledge of the work already done in the field. The results are represented in the following table 1.1 which shows the author, Methods, Results and limitations.

Table 1.1: Summary of Related Work for LearnNova

Author	Method	Results	Limitations
Yuwei Guo et al. [1]	Introduced diffusion-based motion module integration for text-to-video generation.	Demonstrated effective motion conditioning for creating short animation sequences.	Limited to short-duration outputs with constrained temporal dynamics.
Haoxin Chen et al. [2]	Utilized spatio-temporal adapters with high-quality diffusion frameworks.	Provided a robust reference for large-scale video synthesis with strong visual coherence.	Required extensive computational resources for training and rendering.
Zhuoyi Yang et al. [3]	Proposed transformer-diffusion hybrid architectures for long-term video generation.	Showed improved semantic continuity and contextual preservation in generated videos.	Depended heavily on large-scale compute resources and pretraining datasets.
Weijie Kong et al. [4]	Integrated multimodal long-sequence modeling with diffusion-based training.	Achieved enhanced synchronization between video and text modalities.	Incurred high computational cost and limited real-time scalability.
Li et al. [5]	Proposed a two-stage agentic framework using LLMs to create Manim-based theorem animations.	Enabled symbolic reasoning and visual explanations for mathematical theorems.	Limited in synchronization and domain generalization across topics.
Herrmann et al. [6]	Designed an automated system to convert scientific text into Manim animations.	Simplified visualization of abstract research and academic concepts.	Lacked synchronized narration and fine-grained animation control.

Author	Method	Results	Limitations
Yipeng Yu et al. [7]	Implemented a text-to-video pipeline combining narration synthesis and animation rendering.	Demonstrated multimodal storytelling through LLM-guided video synthesis.	Focused primarily on narrative videos rather than instructional material.
Mingyang Mao et al. [8]	Developed a multimodal retrieval-augmented generation pipeline for educational video creation.	Combined diverse data inputs for accurate, AI-driven educational content generation.	Suffered from limited domain-specific fine-tuning and rendering fidelity.

1.4 Conclusion

The literature indicates that video generation has evolved from handcrafted visual synthesis to fully automated generative pipelines based on diffusion and transformer architectures. While these frameworks achieve unprecedented realism and fluency, they remain primarily focused on artistic and cinematic applications. Educational content generation presents distinct challenges requiring conceptual accuracy, coherence, and accessibility rather than aesthetic quality alone. **LearnNova** emerges as a response to this gap, synthesizing insights from existing video generators into a specialized framework that automates the creation of mathematically and physically accurate tutorial videos using large language models, text-to-speech narration, and fine-tuned Manim-based animation.

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